Adaptive Backstepping Control of a Quadcopter With Uncertain Vehicle Mass, Moment of Inertia, and Disturbances

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***Abstract*—In this article, we propose a solution to the problem of path following for a quadcopter aircraft with unknown vehicle parameters (mass and moment of inertia) and external disturbances. By employing the backstepping technique, the proposed adaptive control strategy guarantees the following: the quadcopter is globally steered toward, and kept within, an arbitrarily small neighborhood of a desired smooth path, achieving global uniformly ultimately boundedness; compared to trajectory tracking, a smoother convergence is obtained as the control actuation signals (thrust force and torque) are bounded with respect to the position error, and the designed timing law ensures that the desired path starts to move only when the vehicle gets close to the desired path; and a single adaptive control law can be used for accurate motion control of aerial vehicles with a wide range of inertial properties, without the need for retuning control gains or other parameters. Moreover, the controller is also made robust to external constant and slowly time-varying disturbances through the design of disturbance estimators. To demonstrate the effectiveness and performance of the proposed control strategies, simulation and experimental results are presented and analyzed.**

***Index Terms*—Disturbances, estimator design, model uncertainties, quadcopter aircraft, robust flight control.**

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I. Introduction

FOR the past several decades, flight control of unmanned aerial vehicles (UAVs) has received substantial research interest in view of their auspicious applications in both civilian and military areas, which range from aerial mapping, disaster monitoring, power-line inspection, precision agriculture, target tracking, delivery, to name a few [1]–[5]. Particularly, when compared with fixed-wing aerial vehicles, unmanned rotorcraft present some distinctive features, for example, vertical takeoff-and-landing and hovering, making them ideal test platforms in academic research. The objective of this article is to propose a solution to the problem of steering a quadcopter aircraft to follow a desired path with unknown vehicle mass, moment of inertia, and external disturbances.

Developing flight control strategies for a single UAV is challenging due to its *underactuated* nature, inherent dynamic nonlinearities, and external disturbances. The underactuated nature makes the quadcopter not satisfy Brockett's necessary condition [6] for the existence of stabilizing smooth time-invariant feedback controllers. Various strategies have been investigated and proposed for the purpose of stabilizing the UAV along a reference trajectory (*trajectory tracking task*), ranging from linear control methods, including PID [7] and linear–quadratic regulator (LQR) [8], to nonlinear control techniques, e.g., geometric control [9], [10], sliding mode control (SMC) [11]–[13], backstepping [14]–[17], nonlinear model predictive control (NMPC) [18], [19], neural networks (NN), and fuzzy logic system (FLS)-based controllers [20], [21].

The aforementioned PID and LQR controllers have demonstrated good performance in steering a quadcopter vehicle to track a straight line or trimming trajectory. However, both of them rely on system linearization, leading to the local stability of the closed-loop system and performance degradation in complex operation environment where inevitable uncertainties, such as aerodynamic disturbances, exist [22]. Nonlinear tracking control strategies, such as the ones presented in [9], [12], [20], and [21], can provide larger regions of attraction with disturbances rejection capability, as detailed in the sequel. The geometric tracking controller designed in [9] enabled a quadrotor unmanned aerial vehicle to perform complex flight maneuvers, achieving uniformly ultimately bounded stability. Building on SMC's features, a good robustness performance in the presence of model uncertainties and external disturbances, in [12], the authors developed a sliding mode tracking controller for a helicopter. Stability proof and simulation results presented therein validating that the vehicle can track the desired attitude and position. However, neither [9] or [12] presented experimental results to further validate the effectiveness of the proposed control methods therein. Notice that, in [19], an NMPC trajectory tracking controller was proposed for rotary wing microaerial vehicles, and experimental results were provided as well. But, the stability of the closed-loop system was not proved. Even though NN and FLS presented in [20] and [21] have the ability to approximate an unknown continuous function, to our best knowledge, it is still not clear how many FLS rules/neurons are required to reduce the estimation error to a specific value. In this case, theoretically, the ultimate upper bounds of error terms cannot be defined precisely. Moreover, the reader is referred to [23]–[25] for robust controllers with application to a class of nonlinear systems, and [26]–[28] for different MPC controllers.

A common concern among the mentioned works (*solutions to the trajectory tracking problem*) is the existence of temporal constraints in trajectory tracking task that could cause problems like jerky motions or large actuations. To relax these constraints and achieve a smoother convergence to the desired path, a path-following task is considered instead that can be separated into two subtasks, the geometric (*driving the vehicle to the desired path*) and dynamic task (*getting the vehicle to the desired velocity*).

Numerous solutions to the path-following task have been reported, among them, we highlight [29]–[31] for their application to rotorcraft. In [29], a nonlinear path-following controller was proposed to steer a model-scaled autonomous helicopter along a predefined path while tracking a desired velocity profile. However, the solution reported in [29] can only ensure the path-following errors locally ultimately bounded. To overcome this restriction, a nonlinear global path-following controller was developed in [30], guaranteeing global convergence of the closed-loop path-following error to zero, but, the presented controller design procedure therein suffered from the computation complexity problem. In addition, the authors of [31] proposed a stable path-following controller while the energetic cost of flying was addressed. However, the proposed methodology relied on a simplified kinematic model of a quadrotor, posing the degradation performance problem once the vehicle was operated in a complex environment.

Notice that all aforestated solutions to the trajectory tracking and path-following problems require that the vehicle's mass and moment of inertia are known. Model dynamic uncertainties, when considered, include mainly uncertain aerodynamic damping coefficients. In some specific application scenarios, e.g., load transportation, different loads with unknown diverse weights and sizes could result in large errors. Additionally, for a wide range of quadcopters, their corresponding vehicle masses and inertia moments will be different as well. In this case, it is inappropriate to use the aforementioned methods to control them directly, only if the vehicle masses and inertia moments are tuned to appropriate values (close to their corresponding actual values).

Some adaptive control strategies have been proposed to deal with uncertain vehicle mass and inertia moment, such as the ones presented in [32]–[34]. In [32], the immersion-and-invariance (I&I) approach was proposed to compensate the parametric uncertainties associated with inertia moment. But, the authors still needed to assume that the vehicle's mass was known. Similar approach can also be found in [33], where an adaptive estimating law for the vehicle mass was designed through I&I. However, the authors could not deal with disturbances and unknown mass appearing in the translational motion dynamic model simultaneously. In [34], an adaptive trajectory tracking control strategy for quadrotor microaerial vehicles was proposed, where the vehicle's mass was assumed to be known, but the weight of transported objects was assumed to be unknown. The proposed adaptive strategy therein was mainly used to deal with the disturbances and geometric center noncoincident with the center of mass caused by the added transported objects. The reader is referred to [35] for the robust adaptive controller design for a normal linear multiple-input–multiple-output dynamical system.

In this article, our method derives from the observation that it is useful in practice to have an adaptive control strategy that can be used for the following:

1) steer a wide range of quadcopter vehicles (with different masses and inertia moments) toward, and move along a smooth desired path with a desired speed for aerial mapping, surveillance, etc.;
2) drive a single quadcopter to carry loads (with varied size and weight distributions) smoothly, robustly and accurately.

Motivated by these considerations and the aforementioned studies, the key contributions of this article are as follows.

1) Proposing a nonlinear adaptive path-following controller for an underactuated quadcopter that achieves global uniformly ultimately boundedness (GUUB) in the presence of unknown disturbances and parametric uncertainties.
2) Devising a timing law for the desired path, together with the guarantee that the actuations are bounded with respect to the position error, leading to a smoother convergence compared to that obtained with trajectory tracking.
3) Designing estimators for the quadcopter's mass, external disturbances, and unknown constant terms associated to the quadcopter's moment of inertia, which are explicitly incorporated in the control law to achieve a good robust adaptive path-following performance.

In respect of the state-of-the-art, the main distinctive features of the proposed controller are elaborated as follows. In contrast to the nonlinear Lyapunov-based controller presented in [30] and [36], the proposed control scheme requires one less backstepping iteration, which reduces the order of the feedback system, leading to a simpler controller with better robustness. Unlike the NMPC methods reported in [18] and [19], where the vehicle's model must be exactly known, in this article, uncertain vehicle parameters are considered. Unlike the results in [32]–[34], where only partial uncertain parameters are considered, this article tackles uncertain vehicle mass, moment of inertial, and external disturbances simultaneously.

II. Notation

In this article, $\mathbb{R}^n$ denotes the n-dimensional Euclidean space. A function f is of class $\mathcal{C}^n$ if its derivatives $f,' f,'' \ldots, f^{(n)}$ exist and are continuous. For a constant vector $\mathbf{x} = [x_1, \ldots, x_n]^\mathsf{T}$, the norm is denoted by $||\mathbf{x}|| = \sqrt{\mathbf{x}^\mathsf{T}\mathbf{x}}$, its estimate by $\widehat{\mathbf{x}}$, the estimation error by $\widetilde{\mathbf{x}} = \widehat{\mathbf{x}} - \mathbf{x}$, and its dynamics verify $\dot{\widetilde{\mathbf{x}}} = \dot{\widehat{\mathbf{x}}}$. For a matrix $\Theta \in \mathbb{R}^{n\times n}$, its Frobenius norm is denoted by $||\Theta||_F = (\mathsf{trace}(\Theta^\mathsf{T}\Theta))^{\frac{1}{2}}$. For a diagonal matrix $\mathbf{\Omega} = \mathrm{diag}(\Omega_1, \ldots, \Omega_n)$, $\mathrm{Diag}(\mathbf{\Omega})$ is defined as $\mathrm{Diag}(\mathbf{\Omega}) = [\Omega_1, \ldots, \Omega_n]^\mathsf{T}$. The operator $\circ$ is used to indicate the following multiplication operation, $\mathbf{x} \circ \mathbf{y} = [x_1\mathbf{y}^\mathsf{T}, \ldots, x_n\mathbf{y}^\mathsf{T}]^\mathsf{T}$, where $\mathbf{y} = [y_1, \ldots, y_n]^\mathsf{T}$. Unit vectors $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$ are defined as $\mathbf{e}_1 = [1\ 0\ 0]^\mathsf{T}$, $\mathbf{e}_2 = [0\ 1\ 0]^\mathsf{T}$, and $\mathbf{e}_3 = [0\ 0\ 1]^\mathsf{T}$, $\mathbf{1}_{m\times n} \in \mathbb{R}^{m\times n}$ is a matrix whose entries are all one.

III. Problem Formulation

A. Vehicle Modeling

The quadcopter vehicle is modeled as a rigid body subjected to external forces actuated in thrust and torque. We first introduce an inertial coordinate frame $\{I\}$ and a body-fixed coordinate frame $\{B\}$ attached to the vehicle's center of mass. Notice that the configuration of the body frame with respect to the inertial frame can be viewed as an element of the special Euclidean group, $(\mathbf{R}, \mathbf{p}) = ({}^I_B\mathbf{R}, {}^I\mathbf{p}_B) \in \mathrm{SE}(3)$, where $\mathbf{p} \in \mathbb{R}^3$ represents the position and $\mathbf{R} \in \mathrm{SO}(3)$ is the rotation matrix.

The kinematic equations of motion of the quadcopter are written as

$$\dot{\mathbf{p}} = \mathbf{R}\mathbf{v} \tag{1}$$

$$\dot{\mathbf{R}} = \mathbf{R}\mathbf{S}(\boldsymbol{\omega}) \tag{2}$$

and the dynamic equations

$$\dot{\mathbf{v}} = -\mathbf{S}(\boldsymbol{\omega})\mathbf{v} - m^{-1}T\mathbf{e}_3 + g\mathbf{R}^\mathsf{T}\mathbf{e}_3 + \mathbf{R}^\mathsf{T}\mathbf{b} \tag{3}$$

$$\dot{\boldsymbol{\omega}} = -\mathbf{J}^{-1}\mathbf{S}(\boldsymbol{\omega})\mathbf{J}\boldsymbol{\omega} + \mathbf{J}^{-1}\boldsymbol{\tau} \tag{4}$$

where the map $\mathbf{S}(\cdot) : \mathbb{R}^3 \mapsto \mathbb{R}^{3\times 3}$ yields a skew-symmetric matrix that verifies $\mathbf{S}(\mathbf{x})\mathbf{y} = \mathbf{x} \times \mathbf{y}$ for $\mathbf{x}$ and $\mathbf{y} \in \mathbb{R}^3$. The linear velocity $\mathbf{v} \in \mathbb{R}^3$ and the angular velocity $\boldsymbol{\omega} \in \mathbb{R}^3$ are expressed in the body frame $\{B\}$. The scalar $m \in \mathbb{R}$ and the matrix $\mathbf{J} \in \mathbb{R}^{3\times 3}$ denote the quadcopter's mass and moment of inertia, respectively. The gravitational acceleration is denoted by g. The control inputs include the thrust force $T \in \mathbb{R}$ and torque $\boldsymbol{\tau} \in \mathbb{R}^3$. The disturbances are represented by $\mathbf{b}$, which is assumed to be irrotational. Moreover, the following assumption is made for the quadcopter model.

Assumption 1: The external unknown disturbances $\mathbf{b}$, the quadcopter's mass m in (3), and the moment of inertia in (4), are constant, unknown, but their distribution ranges in

$$\|\mathbf{b}\| \in [0,\, b_{\max}],\ m \in [m_{\min},\ m_{\max}],\ \|\mathbf{J}\|_F \in [J_{\min},\ J_{\max}]$$

where $b_{\max}$, $m_{\min}$, $m_{\max}$, $J_{\min}$, and $J_{\max}$ are known positive numbers. Furthermore, $\mathbf{J}$ is assumed diagonal due to vehicle symmetry.

Disturbances, and in particular, the wind, are never exactly constant. However, this assumption is a convenient and efficient starting point for designing a control method to enhance the stability of the vehicle, even if the disturbances are not necessarily constant. A fitting parallel is the PID controller, where the introduced integral action can not only reject constant disturbances but also enhance the performance of the vehicle for slowly time-varying disturbances. In addition, the constant assumption of the vehicle mass and moment of inertia, as stated in Assumption 1, does not mean that they cannot change. For instance, a single quadcopter is used to execute payload transportation mission: without payload, the system's mass equals to the vehicle's mass; and with payload, the system's mass is the summarization of the vehicle's mass and the payload's mass. During these two operation scenarios, we assume the system's masses are two different constants and both of them stay within $[m_{\min},\ m_{\max}]$.

B. Smooth Projection

Following [37] and [38], we introduce a smooth nondecreasing function $\pi(s)$ given by

$$\pi(s) = \begin{cases} s_{\max} + \varepsilon\left(1 - \exp\left(\dfrac{s_{\max} - s}{\varepsilon}\right)\right), & s > s_{\max} \\ s, & s \in [s_{\min}, s_{\max}] \\ s_{\min} - \varepsilon\left(1 - \exp\left(\dfrac{s - s_{\min}}{\varepsilon}\right)\right), & s < s_{\min} \end{cases} \tag{5}$$

with the following properties:

1) $\pi(s)$ is of class $\mathcal{C}^1$;
2) $\pi(s) \in [s_{\min} - \varepsilon,\ s_{\max} + \varepsilon]$;

where $s_{\max} > s_{\min}$, and ε is an arbitrarily small positive number. To construct a positive definite function, we define $\widetilde{s} = \widehat{s} - s$ and

$$V(s,\ \widetilde{s}) = \int_0^{\widetilde{s}} (\pi(\varphi + s) - s)d\varphi \tag{6}$$

which is positive definite w.r.t. $\tilde{s}$ for $s \in [s_{\min}, s_{\max}]$ (see [37]). Computing the time derivative of $V(s,\ \widetilde{s})$, we have

$$\dot{V}(s,\ \widetilde{s}) = (\pi(\widehat{s}) - s)\dot{\widehat{s}}.$$

The function $V(s,\ \widetilde{s})$ will be used as part of a composite Lyapunov function for our closed-loop system. In addition, for a vector $\mathbf{s} = [s_1, \ldots, s_n]^\mathsf{T}$, $\pi(\mathbf{s})$ is defined as $\pi(\mathbf{s}) = [\pi(s_1), \ldots, \pi(s_n)]^\mathsf{T}$.

C. Path-Following Problem

Let $\mathbf{p}_d(\gamma) \in \mathbb{R}^3$ be a curve of class at least $\mathcal{C}^4$, parameterized by $\gamma \in \mathbb{R}$, with all its partial derivatives with respect to γ bounded. The control objective of the path-following problem is to design a law for the vehicle actuations, thrust force T, and torque $\boldsymbol{\tau}$, and a timing law for the path parameter γ, which ensures convergence of the vehicle's position $\mathbf{p}(t)$ to an arbitrarily small neighborhood of a desired path $\mathbf{p}_d(\gamma)$.

IV. Control Algorithms Design

In this section, a path-following controller is designed for a quadcopter with uncertain vehicle parameters and unknown external disturbances. The controller is designed following a backstepping-like process, and we depart from a Lyapunov candidate function based on the position error, and introduce Lyapunov candidate functions iteratively, until thrust force, torque actuations, and estimators for unknown quantities are obtained.

A. Actuation and Estimation Laws

Define the position error in the inertial frame as

$$\mathbf{z}_1 = \mathbf{p} - \mathbf{p}_d. \tag{7}$$

To introduce a saturation function for $\mathbf{z}_1$, we consider the following Lyapunov candidate function:

$$V_1 = \sigma_{\max} \ln(\cosh(\mathbf{z}_1))^\mathsf{T} \mathbf{1}_{3\times 1}$$

whose time derivative is

$$\dot{V}_1 = \sigma(\mathbf{z}_1)^\mathsf{T}(\mathbf{R}\mathbf{v} - \dot{\mathbf{p}}_d)$$

where $\sigma(\mathbf{z}_1) = \sigma_{\max}\tanh(\mathbf{z}_1)$, $\sigma_{\max}$ is a positive constant. In order to create a negative definite term for $\dot{V}_1$, we add and subtract $k_1\sigma(\mathbf{z}_1)^\mathsf{T}\sigma(\mathbf{z}_1)$, resulting in

$$\dot{V}_1 = -W_1(\mathbf{z}_1) + \sigma(\mathbf{z}_1)^\mathsf{T}(\mathbf{R}\mathbf{v} - \dot{\mathbf{p}}_d + k_1\sigma(\mathbf{z}_1)) \tag{8}$$

where $W_1(\mathbf{z}_1) = k_1\sigma(\mathbf{z}_1)^\mathsf{T}\sigma(\mathbf{z}_1)$ is positive definite.

Notice that the quadcopter is an underactuated vehicle since it has fewer control inputs than the degrees of freedom. The underactuated nature makes the quadcopter not satisfy Brockett's necessary condition [6] for the existence of stabilizing smooth time-invariant feedback controllers, and a naive backstepping implementation could lead to singularities in the control law. To avoid such singularity, we take a clue from [39], and drive the velocity error $(\mathbf{R}\mathbf{v} - \dot{\mathbf{p}}_d + k_1\sigma(\mathbf{z}_1))$ to a small constant vector $\boldsymbol{\delta} = [0,\, 0,\, \delta]^\mathsf{T}$, with $\delta \neq 0$, instead of $\mathbf{0}_{3\times 1}$. Then, we define a new error $\mathbf{z}_2$ as

$$\mathbf{z}_2 = \mathbf{R}\mathbf{v} - \dot{\mathbf{p}}_d + k_1\sigma(\mathbf{z}_1) - \mathbf{R}\boldsymbol{\delta} \tag{9}$$

allowing us to rewrite (8) as

$$\dot{V}_1 = -W_1(\mathbf{z}_1) + \sigma(\mathbf{z}_1)^\mathsf{T}(\mathbf{z}_2 + \mathbf{R}\boldsymbol{\delta}). \tag{10}$$

Following the backstepping procedure, we consider a second Lyapunov candidate function incorporating $\mathbf{z}_2$ as

$$V_2 = V_1 + \frac{1}{2}\mathbf{z}_2^\mathsf{T}\mathbf{z}_2$$

with time derivative

$$\begin{aligned}\dot{V}_2 = &-W_1(\mathbf{z}_1) + \sigma(\mathbf{z}_1)^\mathsf{T}(\mathbf{R}\boldsymbol{\delta} + \mathbf{z}_2) + \mathbf{z}_2^\mathsf{T}\mathbf{R}\,(-\,\eta T\mathbf{e}_3 \\ &-\mathbf{S}(\boldsymbol{\omega})\boldsymbol{\delta} + \mathbf{R}^\mathsf{T}(g\mathbf{e}_3 + \mathbf{b} - \ddot{\mathbf{p}}_d + k_1\dot{\sigma}(\mathbf{z}_1)))\end{aligned} \tag{11}$$

where $\eta = m^{-1}$. By adding and subtracting $k_2\mathbf{z}_2^\mathsf{T}\sigma(\mathbf{z}_2)$ on the right-hand side of (11), we get

$$\begin{aligned}\dot{V}_2 = &-W_2(\mathbf{z}_1,\, \mathbf{z}_2) + \sigma(\mathbf{z}_1)^\mathsf{T}(\mathbf{R}\boldsymbol{\delta} + \mathbf{z}_2) + \mathbf{z}_2^\mathsf{T}\mathbf{R} \\ &\times\left(-\eta T\mathbf{e}_3 - \mathbf{S}(\boldsymbol{\omega})\boldsymbol{\delta} + \mathbf{R}^\mathsf{T}\left(g\mathbf{e}_3 + \mathbf{b}\right.\right. \\ &\left.\left.-\,\ddot{\mathbf{p}}_d + k_1\dot{\sigma}(\mathbf{z}_1) + k_2\sigma(\mathbf{z}_2)\right)\right)\end{aligned} \tag{12}$$

where $W_2(\mathbf{z}_1,\, \mathbf{z}_2) = W_1(\mathbf{z}_1) + k_2\mathbf{z}_2^\mathsf{T}\sigma(\mathbf{z}_2)$, and k_2 is a positive control gain.

Notice that the unknown parameters η and $\mathbf{b}$, in (12), cannot be included in the control law. We thereby introduce estimators for those quantities as $\widehat{\eta}$ and $\widehat{\mathbf{b}}$ to be used for feedback. In order to ensure that all these estimates remain within *a priori* bounded set, the smooth projection $\pi(s)$ introduced in Section III-B is used. We now define the new Lyapunov candidate function including the estimate errors as

$$V_{2b} = V_2 + \lambda_\eta^{-1}V(\eta, \widetilde{\eta}) + \sum_{i=1}^{3}\lambda_i^{-1}V(b_i, \tilde{b}_i)$$

where $\lambda_\eta > 0$, $\Lambda_\mathbf{b} = \mathrm{diag}(\lambda_1,\, \lambda_2,\, \lambda_3)$ is a positive definite diagonal matrix, and b_i are the components of the disturbances $\mathbf{b}$, and $V(s,\, \tilde{s})$ is defined in (6), which is used for introducing $\pi(s)$. Computing the time derivative of V_{2b}, we have

$$\begin{aligned}\dot{V}_{2b} = &-W_2(\mathbf{z}_1,\, \mathbf{z}_2) + \sigma(\mathbf{z}_1)^\mathsf{T}(\mathbf{R}\boldsymbol{\delta} + \mathbf{z}_2) + \mathbf{z}_2^\mathsf{T}\mathbf{R} \\ &\times\left(-\pi(\widehat{\eta})T\mathbf{e}_3 + \mathbf{e}_3\mathbf{e}_3^\mathsf{T}(-\mathbf{S}(\boldsymbol{\omega})\boldsymbol{\delta} + \boldsymbol{\zeta})\right. \\ &\left.+\,(\mathbf{I}_{3\times 3} - \mathbf{e}_3\mathbf{e}_3^\mathsf{T})(-\mathbf{S}(\boldsymbol{\omega})\boldsymbol{\delta} + \boldsymbol{\zeta})\right) \\ &+(\pi(\widehat{\eta}) - \eta)\left(\mathbf{z}_2^\mathsf{T}\mathbf{R}T\mathbf{e}_3 + \lambda_m^{-1}\dot{\widehat{\eta}}\right) \\ &+(\boldsymbol{\pi}(\widehat{\mathbf{b}}) - \mathbf{b})^\mathsf{T}\left(-\mathbf{z}_2 + \Lambda_\mathbf{b}^{-1}\dot{\widehat{\mathbf{b}}}\right)\end{aligned} \tag{13}$$

where

$$\boldsymbol{\zeta} = \mathbf{R}^\mathsf{T}\left(g\mathbf{e}_3 - \ddot{\mathbf{p}}_d + k_1\dot{\sigma}(\mathbf{z}_1) + k_2\sigma(\mathbf{z}_2) + \boldsymbol{\pi}(\widehat{\mathbf{b}})\right).$$

In order to zero out the last two terms of (13), we set the estimation laws for $\dot{\widehat{\eta}}$ and $\dot{\widehat{\mathbf{b}}}$ as

$$\dot{\widehat{\eta}} = -\lambda_m\mathbf{z}_2^\mathsf{T}\mathbf{R}T\mathbf{e}_3 \tag{14}$$

$$\dot{\widehat{\mathbf{b}}} = \Lambda_\mathbf{b}\mathbf{z}_2. \tag{15}$$

To cancel $(\mathbf{e}_3\mathbf{e}_3^\mathsf{T}(-\mathbf{S}(\boldsymbol{\omega})\boldsymbol{\delta} + \boldsymbol{\zeta}))$, the thrust force T is set as

$$T = \pi(\widehat{\eta})^{-1}\mathbf{e}_3^\mathsf{T}(-\mathbf{S}(\boldsymbol{\omega})\boldsymbol{\delta} + \boldsymbol{\zeta}) \tag{16}$$

where $\pi(\widehat{\eta}) \in [m_{\max}^{-1} - \varepsilon,\, m_{\min}^{-1} + \varepsilon]$, and ε is chosen such that $m_{\max}^{-1} > \varepsilon > 0$, guaranteeing that $\pi(\widehat{\eta})$ is always invertible. It is important to point out that this is the reason why we need to introduce and use the projection function $\pi(s)$, otherwise, singularity could arise since we cannot ensure that $\hat{\eta} > 0$ always hold. Substituting (14)–(16) into (13), we obtain

$$\begin{aligned}\dot{V}_{2b} = &-W_2(\mathbf{z}_1,\, \mathbf{z}_2) + \sigma(\mathbf{z}_1)^\mathsf{T}(\mathbf{R}\boldsymbol{\delta} + \mathbf{z}_2) \\ &+(\Pi_{12}\mathbf{R}^\mathsf{T}\mathbf{z}_2)^\mathsf{T}(\boldsymbol{\delta}_n\boldsymbol{\omega} + \Pi_{12}\boldsymbol{\zeta})\end{aligned} \tag{17}$$

where Π_{12} and $\boldsymbol{\delta}_n$ are given by

$$\Pi_{12} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad \boldsymbol{\delta}_n = \begin{bmatrix} 0 & -\delta & 0 \\ \delta & 0 & 0 \end{bmatrix}.$$

To cancel the last term of (17), i.e., $(\Pi_{12}\mathbf{R}^\mathsf{T}\mathbf{z}_2)^\mathsf{T}(\boldsymbol{\delta}_n\boldsymbol{\omega} + \Pi_{12}\boldsymbol{\zeta})$, we choose the virtual desired angular velocity as

$$\boldsymbol{\omega}_d = -\boldsymbol{\delta}_m\boldsymbol{\zeta}$$

where $\boldsymbol{\delta}_m = \boldsymbol{\delta}_n^\mathsf{T}(\boldsymbol{\delta}_n\boldsymbol{\delta}_n^\mathsf{T})^{-1}\Pi_{12}$, which is always well defined since the determinant $\det(\boldsymbol{\delta}_n\boldsymbol{\delta}_n^\mathsf{T}) = \delta^4$ is nonzero for $\delta \neq 0$.

To continue the backstepping, we define the angular velocity error $\mathbf{z}_3$ as

$$\mathbf{z}_3 = \boldsymbol{\omega} - \boldsymbol{\omega}_d \tag{18}$$

that one would like to drive to zero. Correspondingly, a new Lyapunov candidate function by augmenting $\mathbf{z}_3$ is defined as

$$V_3 = V_{2b} + \frac{\beta}{2}\mathbf{z}_3^\mathsf{T}\mathbf{J}\mathbf{z}_3$$

where $\beta > 0$. Computing the time derivative of V_3, in closed loop, we have

$$\begin{aligned}\dot{V}_3 = -W_3(\mathbf{z}_1,\, \mathbf{z}_2,\, \mathbf{z}_3) + \sigma(\mathbf{z}_1)^\mathsf{T}(\mathbf{R}\boldsymbol{\delta} + \mathbf{z}_2) + \beta\mathbf{z}_3^\mathsf{T}\Big(\boldsymbol{\tau} \\ - \mathbf{S}(\boldsymbol{\omega})\mathbf{J}\boldsymbol{\omega} + \mathbf{J}\boldsymbol{\delta}_m\dot{\boldsymbol{\zeta}} + \beta^{-1}\left(\boldsymbol{\delta}_n^\mathsf{T}\Pi_{12}\mathbf{R}^\mathsf{T}\mathbf{z}_2 + k_3\sigma(\mathbf{z}_3)\right)\Big)\end{aligned}$$

where $W_3(\mathbf{z}_1,\, \mathbf{z}_2,\, \mathbf{z}_3) = W_2(\mathbf{z}_1,\, \mathbf{z}_2) + k_3\mathbf{z}_3^\mathsf{T}\sigma(\mathbf{z}_3)$ is positive definite.

At this point, the actuation $\boldsymbol{\tau}$ is available, but, unfortunately, there also exist unknown terms $\mathbf{J}$, η, and $\mathbf{b}$ in $(-\mathbf{S}(\boldsymbol{\omega})\mathbf{J}\boldsymbol{\omega} + \mathbf{J}\boldsymbol{\delta}_m\dot{\boldsymbol{\zeta}})$. By direct expansion, we have $-\mathbf{S}(\boldsymbol{\omega})\mathbf{J}\boldsymbol{\omega} + \mathbf{J}\boldsymbol{\delta}_m\dot{\boldsymbol{\zeta}} = \mathbf{W}\mathbf{w}$, where $\mathbf{W}$ is a state dependent matrix and $\mathbf{w}$ is the constant vector of unknowns

$$\mathbf{w} = [\mathrm{Diag}(\mathbf{J})^\mathsf{T},\, \eta\,\mathrm{Diag}(\mathbf{J})^\mathsf{T},\, (\mathrm{Diag}(\mathbf{J}) \circ \mathbf{b})^\mathsf{T}]^\mathsf{T}.$$

To get the control input $\boldsymbol{\tau}$, similar to what we did for η and $\mathbf{b}$ in (12), here, we also introduce estimations for $\mathbf{w}$, denoted as $\widehat{\mathbf{w}}$. Then, the Lyapunov derivative $\dot{V}_3$ can be rewritten as

$$\begin{aligned}\dot{V}_3 = &-W_3(\mathbf{z}_1,\, \mathbf{z}_2,\, \mathbf{z}_3) + \sigma(\mathbf{z}_1)^\mathsf{T}(\mathbf{R}\boldsymbol{\delta} + \mathbf{z}_2) \\ &+ \beta\mathbf{z}_3^\mathsf{T}\Big(\boldsymbol{\tau} + \mathbf{W}\widehat{\mathbf{w}} + \beta^{-1}\Big(\boldsymbol{\delta}_n^\mathsf{T}\Pi_{12}\mathbf{R}^\mathsf{T}\mathbf{z}_2 \\ &+ k_3\sigma(\mathbf{z}_3)\Big)\Big) - \beta\mathbf{z}_3^\mathsf{T}\mathbf{W}\widetilde{\mathbf{w}}\end{aligned}$$

where $\widehat{\mathbf{w}} = \widehat{\mathbf{w}} - \mathbf{w}$ denotes the estimation error.

Complementing the Lyapunov candidate function V_3 with the estimation errors allows us to devise a stable estimator for the vector of unknowns $\mathbf{w}$ and a stabilizing feedback control $\boldsymbol{\tau}$. Let the final Lyapunov function be

$$V_{3b} = V_3 + \frac{1}{2}\widetilde{\mathbf{w}}^\mathsf{T}\Lambda_\mathbf{w}^{-1}\widetilde{\mathbf{w}}$$

with $\Lambda_\mathbf{w}$ is a positive definite matrix. Computing its time derivative results in

$$\begin{aligned}\dot{V}_{3b} = &-W_3(\mathbf{z}_1,\, \mathbf{z}_2,\, \mathbf{z}_3) + \sigma(\mathbf{z}_1)^\mathsf{T}(\mathbf{R}\boldsymbol{\delta} + \mathbf{z}_2) \\ &+ \beta\mathbf{z}_3^\mathsf{T}\Big(\boldsymbol{\tau} + \mathbf{W}\widehat{\mathbf{w}} + \beta^{-1}\Big(\boldsymbol{\delta}_n^\mathsf{T}\Pi_{12}\mathbf{R}^\mathsf{T}\mathbf{z}_2 \\ &+ k_3\sigma(\mathbf{z}_3)\Big)\Big) + \widetilde{\mathbf{w}}^\mathsf{T}\left(-\beta\mathbf{W}^\mathsf{T}\mathbf{z}_3 + \Lambda_\mathbf{w}^{-1}\dot{\widehat{\mathbf{w}}}\right)\end{aligned} \tag{19}$$

where the stabilizing options for the torque input and estimator dynamics are now clear. Designing the torque input as

$$\boldsymbol{\tau} = -\mathbf{W}\widehat{\mathbf{w}} - \beta^{-1}\left(\boldsymbol{\delta}_n^\mathsf{T}\Pi_{12}\mathbf{R}^\mathsf{T}\mathbf{z}_2 + k_3\sigma(\mathbf{z}_3)\right) \tag{20}$$

so as to zero out $(\beta\mathbf{z}_3^\mathsf{T}(\boldsymbol{\tau} + \mathbf{W}\widehat{\mathbf{w}} + \beta^{-1}(\boldsymbol{\delta}_n^\mathsf{T}\Pi_{12}\mathbf{R}^\mathsf{T}\mathbf{z}_2 + k_3\sigma(\mathbf{z}_3))))$, and the estimator update law as

$$\dot{\widehat{\mathbf{w}}} = \beta\Lambda_\mathbf{w}\mathbf{W}^\mathsf{T}\mathbf{z}_3 \tag{21}$$

such that $\widetilde{\mathbf{w}}^\mathsf{T}(-\beta\mathbf{W}^\mathsf{T}\mathbf{z}_3 + \Lambda_\mathbf{w}^{-1}\dot{\widehat{\mathbf{w}}})$ is canceled.

Substituting (20) and (21) into (19), we have the closed-loop Lyapunov time derivative

$$\dot{V}_{3b} = -W_3(\mathbf{z}_1,\, \mathbf{z}_2,\, \mathbf{z}_3) + \sigma(\mathbf{z}_1)^\mathsf{T}(\mathbf{R}\boldsymbol{\delta} + \mathbf{z}_2)$$

where $W_3(\mathbf{z}_1,\, \mathbf{z}_2,\, \mathbf{z}_3) = k_1\sigma(\mathbf{z}_1)^\mathsf{T}\sigma(\mathbf{z}_1) + k_2\mathbf{z}_2^\mathsf{T}\sigma(\mathbf{z}_2) + k_3\mathbf{z}_3^\mathsf{T}\sigma(\mathbf{z}_3)$, $k_1,\ k_2,$ and k_3 are positive control gains.

B. Timing Law

Let $\dot{\gamma}_d$ denote the desired speed assignment for $\dot{\gamma}$, which is defined as $\dot{\gamma}_d = v_d\psi(||\mathbf{z}_1||)$, where $\psi(\cdot)$ is a smooth bump-like function (of class $\mathcal{C}^\infty$) given by

$$\psi(s) = \begin{cases} 1, & 0 \le |s| < \alpha_1 \\ \dfrac{1}{2}\tanh\left(\dfrac{\theta(s)}{\theta^2(s) - 1}\right) + \dfrac{1}{2}, & \alpha_1 \le |s| \le \alpha_2 \\ 0, & \alpha_2 < |s| \end{cases}$$

and $\theta(s) = (2s - \alpha_1 - \alpha_2)/(\alpha_2 - \alpha_1)$, and α_1 and α_2 are design parameters with $\alpha_2 > \alpha_1 > 0$. This function is identically one for $|s| < \alpha_1$ and zero for $|s| > \alpha_2$. This choice of $\dot{\gamma}_d$ allows for a smoother transient response in path following. That is, when $\|\mathbf{z}_1\|$ is large ($\|\mathbf{z}_1\| > \alpha_2$), the desired path will wait for the quadcopter to approach it. Only when $\|\mathbf{z}_1\| \le \alpha_2$, the desired path starts to move. When $\|\mathbf{z}_1\| \le \alpha_1$, the desired path progresses at a constant speed v_d. In what follows, we are going to design a timing law for $\ddot{\gamma}$ that guarantees the convergence of $\dot{\gamma}$ to $\dot{\gamma}_d$. Define the along-path speed tracking error z_4 as

$$z_4 = \dot{\gamma} - \dot{\gamma}_d \tag{22}$$

and its corresponding Lyapunov candidate function

$$V_4 = \frac{1}{2}z_4^2 \tag{23}$$

with time derivative

$$\dot{V}_4 = -k_4z_4^2 + z_4(\ddot{\gamma} - \ddot{\gamma}_d + k_4z_4) \tag{24}$$

where k_4 is a positive number. Choosing the timing law for $\ddot{\gamma}$ as

$$\ddot{\gamma} = \ddot{\gamma}_d - k_4z_4. \tag{25}$$

Substituting (25) into (24), we get

$$\dot{V}_4 = -k_4z_4^2 \tag{26}$$

which is negative definite.

The path-following result is stated in the forthcoming subsection.

C. Theorem and Stability Analysis

Theorem 1: Let the quadcopter's kinematics and dynamics be described by (1)–(4), and $\mathbf{p}_d(\gamma) \in \mathcal{C}^4$ be the desired path whose partial derivatives with respect to γ are bounded, and γ is a parameter to be designed. Under Assumption 1, and consider the control laws (16) and (20), the estimation laws (14), (15), and (21), and the timing law (25), where $\pi(\cdot)$ is the smooth projection function defined in (5), and $\sigma_{\max}$ satisfies $\sigma_{\max} > ||\boldsymbol{\delta}||/k_1$. Then, for any initial position, the position error $\|\mathbf{z}_1\|$, defined in (7), is globally uniformly ultimately bounded by $\sigma^{-1}(\|\boldsymbol{\delta}\|/k_1)$, while $\|\mathbf{z}_2\|$, $\|\mathbf{z}_3\|$, and $\|z_4\|$, defined in (9), (18), and (22), respectively, converge to zero as time goes to infinity, achieving GUUBs.

Proof: We start the proof by investigating the stability of the $(\mathbf{z}_2,\, \mathbf{z}_3)$-subsystem. Under Assumption 1, we consider a new Lyapunov candidate function

$$V_n = \frac{1}{2}\mathbf{z}_2^{\mathsf{T}}\mathbf{z}_2 + \frac{\beta}{2}\mathbf{z}_3^{\mathsf{T}}\mathbf{J}\mathbf{z}_3 + \frac{1}{2}\widetilde{\mathbf{w}}^{\mathsf{T}}\Lambda_{\mathbf{w}}^{-1}\widetilde{\mathbf{w}} + \lambda_\eta^{-1}V(\eta, \widetilde{\eta}) + \mathrm{diag}(\Lambda_{\mathbf{b}}^{-1})^{\mathsf{T}}V(\mathbf{b}, \widetilde{\mathbf{b}})$$

whose time derivative yields

$$\dot{V}_n = -k_2\mathbf{z}_2^{\mathsf{T}}\sigma(\mathbf{z}_2) - k_3\mathbf{z}_3^{\mathsf{T}}\sigma(\mathbf{z}_3)$$

which is strictly negative definite. As a consequence, V_n is bounded, and it follows that $\mathbf{z}_2$, $\mathbf{z}_3$, $\widetilde{\mathbf{w}}$, $\tilde{\eta}$, and $\tilde{\mathbf{b}}$ are also bounded. Moreover, we can also obtain that $\ddot{V}_n$ is bounded. It follows that $\dot{V}_n$ is uniformly continuous. By using the Barbalat's lemma, we have $\dot{V}_n$ converges to zero. As a consequence, $\mathbf{z}_2$ and $\mathbf{z}_3$ converge to zero as time approaches infinity.

Now, we go back to the $\mathbf{z}_1$-subsystem and consider the upper bound of (10) as

$$\dot{V}_1 \le -||\sigma(\mathbf{z}_1)||\,(k_1\sigma(||\mathbf{z}_1||) - ||\boldsymbol{\delta}|| - ||\mathbf{z}_2||)$$

which is strictly negative definite for $\sigma(\|\mathbf{z}_1\|) > k_1^{-1}(\|\boldsymbol{\delta}\| + \|\mathbf{z}_2\|)$. As we established before, $\mathbf{z}_2$ converges to zero as time goes to infinity, we have $\|\mathbf{z}_1\|$ is globally ultimately bounded by $\sigma^{-1}(\|\boldsymbol{\delta}\|/k_1)$ under the condition that $\sigma_{\max} > \|\boldsymbol{\delta}\|/k_1$. To prove that $\|z_4\|$ is driven to zero, we combine (23) and (26), to obtain $\dot{V}_4 = -2k_4V_4$, from which it follows that $\|z_4\| = e^{-k_4 t}\|z_4(0)\|$. Therefore, $\|z_4\|$ converges to zero exponentially as time approaches infinity. ■

Remark 1: From the proof of Theorem 1, we conclude that larger control gain k_1 and smaller $\boldsymbol{\delta}$ lead to a smaller final position error. It seems reasonable then to always choose a high gain k_1 and small $\boldsymbol{\delta}$, as long as $\delta \neq 0$. However, the thrust force T is proportional to k_1, and the torque $\boldsymbol{\tau}$ relies on δ^{-4}. As a consequence, too large k_1 or too small $\boldsymbol{\delta}$ could cause undesirable fast oscillations of the actuation and system states, even though the error remains bounded. Therefore, a tradeoff is needed between the path-following accuracy and the magnitude of these oscillations.

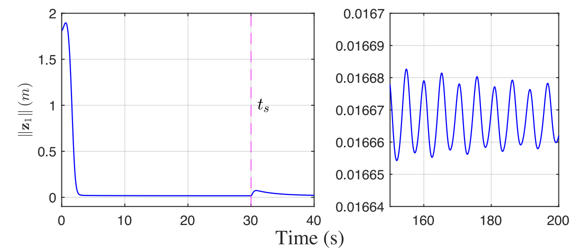

Fig. 1. Time evolution of $||\mathbf{z}_1||$ in a simulation run where the quadrotor mass is abruptly changed at $t = 30$ s. The error norm is ultimately bounded by $\sigma^{-1}(||\boldsymbol{\delta}||/k_1) = 0.0167(\mathrm{m})$.

V. Simulation Results

To attest the performance of the proposed controller, this section presents simulation results. Numerical parameters used for simulation are given as: $k_1 = 4$, $k_2 = 4$, $k_3 = 13$, $k_4 = 1$, $\lambda_m = 0.3$, $\alpha_1 = 0.5$, $\alpha_2 = 1.5$, $\delta = 0.1$, $\mathbf{b} = [0.3,\ 0.2,\ 0.1]^{\mathsf{T}}$, $\sigma_{\max} = 1.5$, and $\psi_d = 0$. The desired path is a tilted *eight-shape* path, which is defined as

$$\mathbf{p}_d(\gamma) = \mathbf{R}_x(\pi/20)\begin{bmatrix}\ell\sin(w\gamma)\cos(w\gamma)\\ \ell\sin(w\gamma)\\ -1.2\end{bmatrix}(\mathrm{m}) \tag{27}$$

where $\ell = 1.2$, $w = 0.6$, and

$$\mathbf{R}_x(\pi/20) = \begin{bmatrix}1 & 0 & 0\\ 0 & \cos(\pi/20) & \sin(\pi/20)\\ 0 & -\sin(\pi/20) & \cos(\pi/20)\end{bmatrix}.$$

To demonstrate the adaptivity of the proposed control method, the simulation is divided into the following two phases:

1) before t_s, the vehicle has a constant mass $m = 0.20\,(\mathrm{kg})$ and constant moment of inertia $\mathbf{J} = \mathrm{diag}(6.5,\ 6.5,\ 13.0)\cdot 10^{-4}\,(\mathrm{kg}\cdot\mathrm{m}^2)$;
2) at t_s, the mass and moment of inertia are changed to $m = 0.245\,(\mathrm{kg})$ and $\mathbf{J} = \mathrm{diag}(7.96,\ 7.96,\ 15.92)\cdot 10^{-4}\,(\mathrm{kg}\cdot\mathrm{m}^2)$, respectively, simulating an increased payload, and remain constants onwards.

The sudden change of mass and moment of inertia can be viewed as a "reset" of the controller. These parameters changed but if they remain constant thereafter, then we have proof that the path will be tracked and the closed-loop system is stable.

The time evolution of $\|\mathbf{z}_1\|$ is displayed in Fig. 1, from where we can see that $\|\mathbf{z}_1\|$ is ultimately bounded by $\sigma^{-1}(\|\boldsymbol{\delta}\|/k_1) = 0.0167\,(\mathrm{m})$, which is the theoretical bound obtained from Theorem 1. Notice that at $t_s = 30\,(\mathrm{s})$, path-following errors undergo a new transient due to the change of vehicle mass and moment of inertia. To further validate and illustrate that the proposed control strategy is able to render the closed-loop system stable, we consider the following.

1) Multiple different initial positions: $\mathbf{p}(0) = [5,\ 5,\ -5]^{\mathsf{T}}\,(\mathrm{m})$, $\mathbf{p}(0) = [15,\ -15,\ 15]^{\mathsf{T}}\,(\mathrm{m})$, and $\mathbf{p}(0) = [-30,\ 30,\ -30]^{\mathsf{T}}\,(\mathrm{m})$.
2) Different added masses: $m_a = 0.1\,(\mathrm{kg})$ and $m_a = 0.2\,(\mathrm{kg})$.

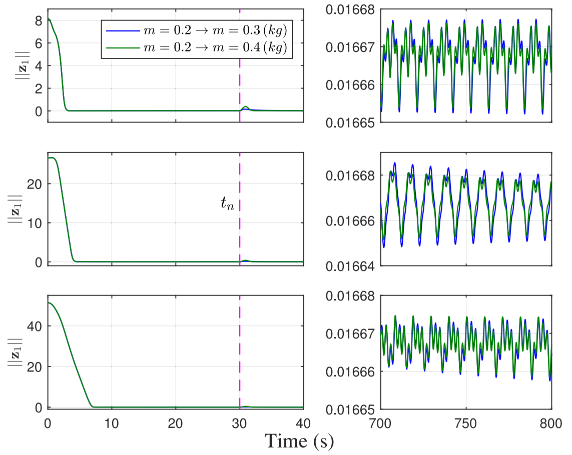

Fig. 2. Time evolution of $\|\mathbf{z}_1\|$ for simulations with different initial position errors $\|\mathbf{z}_1(0)\| = 8.15\,(\mathrm{m})$, $\|\mathbf{z}_1(0)\| = 26.58\,(\mathrm{m})$, and $\|\mathbf{z}_1(0)\| = 51.40\,(\mathrm{m})$.

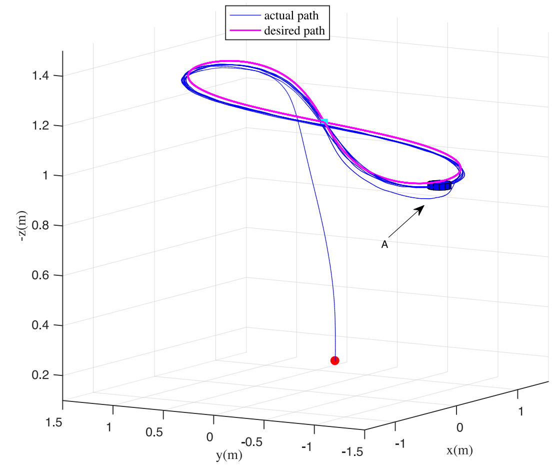

Fig. 3. Time evolution of the actual quadcopter position and the desired path, where • and ■ denote the initial positions of quadcopter and the desired path, respectively.

Fig. 2 displays the time evolution of $\|\mathbf{z}_1\|$, showing that, as expected, it converges to a neighborhood of zero for different initial position errors and is ultimately bounded by $\sigma^{-1}(||\boldsymbol{\delta}||/k_1) = 0.0167(\mathrm{m})$. At $t = t_n$, the mass of the vehicle is changed from $0.2\,(\mathrm{kg})$ and $0.4\,(\mathrm{kg})$, however, the proposed controller still can stabilize the quadcopter and maintain the performance because the existence of update laws for the vehicle mass and terms involved with moment inertia.

VI. Experimental Setup

In order to experimentally demonstrate the performance of the proposed control strategies, we used a MATLAB/Simulink environment that seamlessly integrates the sensors, the control algorithms developed in Section IV, and the communication with the aerial vehicle. The aerial vehicle used for the experiments is a radio controlled 200QX Blade quadcopter, which is low cost and highly maneuverable, making it an ideal test platform.

Considering the fact that there are no on-board sensors on the quadcopter, the state of the vehicle must be measured through external sensors. In our setup, we use a VICON motion capture system, comprising 12 cameras, together with markers attached to the quadcopter. This motion capture system can locate and measure the positions of the markers accurately, from which the position and orientation of the vehicle are obtained. In our case, the motion capture system provides the state measurements at $100\,\mathrm{Hz}$. The controller is implemented in Simulink and runs in soft real time. The actuation signals are sent to the quadcopter through an RF transmitter at maximum rate the radio link allows $(45\,\mathrm{Hz}, 22\,\mathrm{ms}$ interval).

The quadcopter used herein is controlled through thrust and angular velocity commands. In order to employ the proposed controller, the quadrotor system was augmented with an artificial state, mimicking the angular dynamics of an actual quadcopter, (4), that was simulated in Simulink. It is worth pointing out that if we use the proposed control input, $\boldsymbol{\tau}$, defined in (20), directly, the third component of $\boldsymbol{\omega}$ obtained from (4) is always zero, leaving one more degree of freedom to control the yaw angle. Let ψ_d denote the desired yaw angle, and let the yaw control input be

$$\omega_z = -k_p(\psi - \psi_d) - k_i \int_0^t (\psi - \psi_d)dt \tag{28}$$

where k_p and k_i are positive control gains, and the integral term is used to reject external disturbances, guaranteeing that the yaw of the vehicle can always track the desired yaw angle. Moreover, notice that the designed thrust T and angular velocity $\boldsymbol{\omega}$ are expressed in International System of Units, and need to be converted into radio commands (RC) that can be accepted by the radio control system. For the thrust force and angular velocities, their corresponding accepted RC, denoted by T_{RC} and $\boldsymbol{\omega}_{\mathrm{TC}}$, are in the range $[0, 1]$ and $[-1, 1]$, respectively. The relations between T and T_{RC}, and $\boldsymbol{\omega}$ and $\boldsymbol{\omega}_{\mathrm{RC}}$ were identified from experimental data. The following relations were established $T_{\mathrm{RC}} = (T - 2.3071)/3.138$ and $\boldsymbol{\omega}_{\mathrm{RC}} = -[\omega_x/19,\, \omega_y/19,\, \omega_z/8]^{\mathsf{T}}$.

Remark 2: Since we need to compute the angular velocity through (4), therefore, we still need to know the basic knowledge of $\mathbf{J}$ for doing experimental test. To get $\mathbf{J}$, the vehicle is viewed as a cuboid, and follow the formulas presented in [40], each diagonal component of $\mathbf{J}$ can be computed.

VII. Experimental Results

A. Path-Following and Adaptivity Test

The desired path used for the experimental test is the same as for simulation, defined in (27). Fig. 3 contrasts the desired path and the actual path described by the quadcopter, which eventually follows closely the desired path. The deviation in position marked with "A" is caused by the forced change of the vehicle mass (several coins are laid on the quadcopter while it is flying). Correspondingly, as displayed in Fig. 4, the error $||\mathbf{z}_1||$ undergoes a transient at $t = t_s$. To compensate the added mass, the estimated mass $\widehat{m}$ increases from around $0.19\,(\mathrm{kg})$ to $0.23\,(\mathrm{kg})$ (with the estimation error $|\tilde{m}| = 0.015\,(\mathrm{kg})$), while the thrust force T increases from around $1.9\,(\mathrm{N})$ to $2.3\,(\mathrm{N})$, as

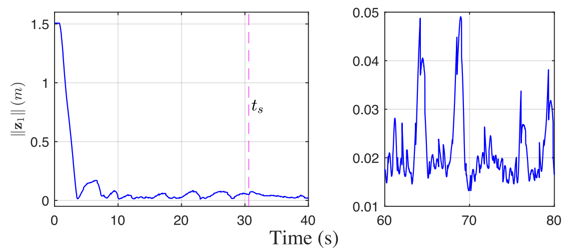

Fig. 4. Time evolution of $||\mathbf{z}_1||$ during a practical experiment using the proposed controller.

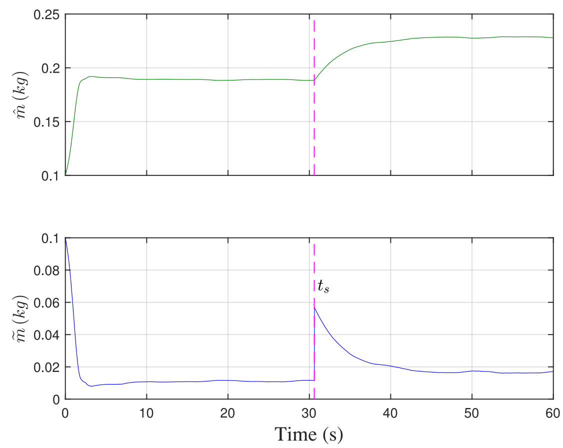

Fig. 5. Time evolution of $\widehat{m}$ and $\widetilde{m}$ during a practical experiment using the proposed controller, with a change of mass forced at around $t = 30$ s.

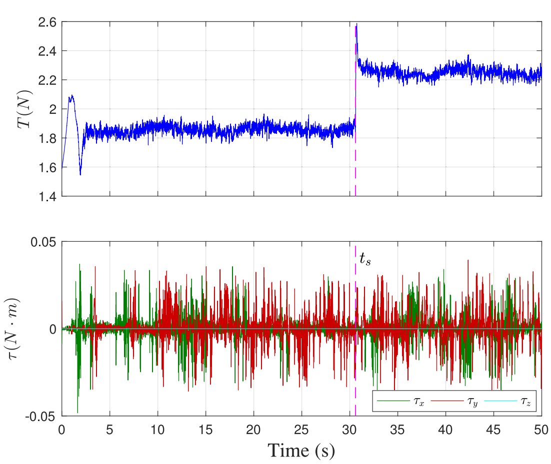

Fig. 6. Time evolution of T and $\boldsymbol{\tau}$ during a practical experiment using the proposed controller, with a change of mass forced at around $t = 30$ s.

shown in Figs. 5 and 6. When $||\mathbf{z}_1||$ returns to a new steady state, its root mean square error (RMSE) $2.3\,(\mathrm{cm})$ and standard deviation (SD) $0.1\,(\mathrm{cm})$.

It is noted that the evolution of the error $||\mathbf{z}_1||$, displayed in Fig. 4, presents similar characteristics to the simulation plot in what regards the convergence rate. However, the state error evolution is more jittery and noisy due to the existence of unavoidable experimental nonidealities such as unmodeled dynamics, imprecise model identification, communication delays, measurement noise, among others. Nonetheless, for most of the time, the steady-state value of the error $||\mathbf{z}_1||$ is around $0.02\,(\mathrm{m})$ and is very close to the theoretical bound of $0.0167\,(\mathrm{m})$ as patent on the right-most plot of Fig. 4.

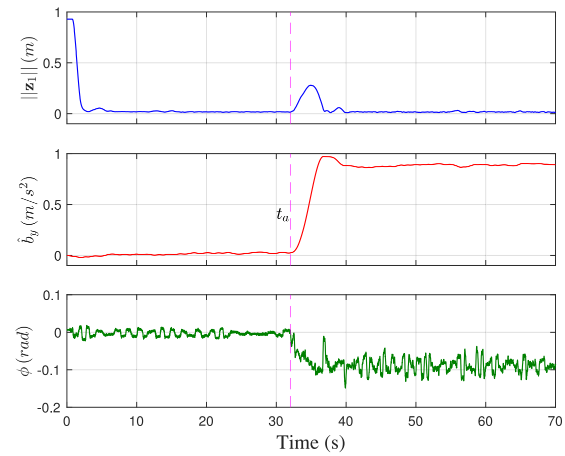

Fig. 7. Time evolution of $||\mathbf{z}_1||$, $\widehat{b}_y$ and roll angle ϕ with disturbance estimators, in the presence of wind generated by a mechanical fan.

B. Robustness to Constant Wind Disturbances

In order to show that the proposed controller is able to reject external constant wind disturbances, we devise a second experiment where the quadcopter is forced to hover at a point $\mathbf{p}_h$ in the presence of wind generated by a mechanical fan. The experimental results with disturbance estimators obtained for the position error, the disturbance estimation, and roll angle are shown in Fig. 7. Before $t = t_a$, the fan is OFF and when the quadcopter reaches a steady state, we have the following:

1) the RMSE and SD of the position error $||\mathbf{z}_1||$ are $1.7\,(\mathrm{cm})$ and $0.1\,(\mathrm{cm})$, respectively;
2) the disturbance estimate $\hat{b}_y$ along the y-axis and the roll angle ϕ are close to zero since there is no wind.

From $t = t_a$ onward, the fan is ON, the value of $||\mathbf{z}_1||$ increases first, then decreases due to the contribution of the disturbance estimators ($\hat{b}_y$ increases initially but finally converges to the neighborhood of $0.9\,(\mathrm{m/s^2})$), while the quadcopter tilts around $-0.1\,(\mathrm{rad})$ so as to compensate for the wind. When the vehicle reaches a new steady state, the RMSE and SD of $||\mathbf{z}_1||$ are $1.7\,(\mathrm{cm})$ and $0.3\,(\mathrm{cm})$, correspondingly.

Fig. 8 depicts the time evolution of $||\mathbf{z}_1||$ using the proposed controller but without the disturbance estimators. At $t = t_b$, the fan is turned ON. When the vehicle eventually reaches a steady state, the position error has RMSE $0.386\,(\mathrm{m})$ and SD $0.206\,(\mathrm{m})$, values much larger than those obtained when the test is performed with the disturbance estimators included. The disturbance estimators, therefore, improve the rejection of the external constant wind and the robust performance of the closed-loop error system is attained.

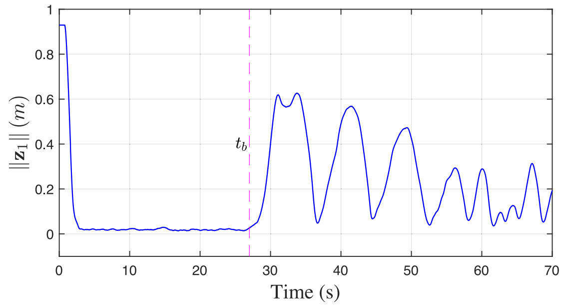

Fig. 8. Time evolution of $||\mathbf{z}_1||$ without disturbance estimators, in the presence of wind generated by a mechanical fan.

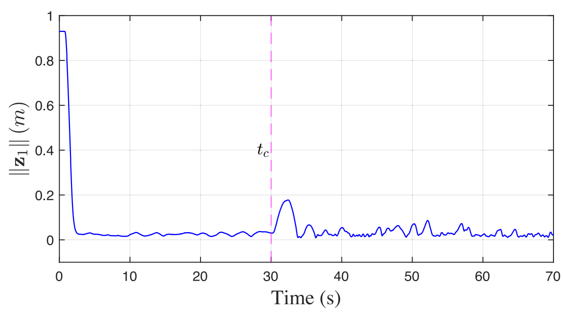

Fig. 9. Time evolution of $||\mathbf{z}_1||$ with disturbance estimators, in the presence of slowly time-varying wind. Even though they are time varying, the disturbances are well mitigated by the proposed control method.

C. Robustness to Time-Varying Wind Disturbances

To further validate that the proposed controller can also reject, and is robust to, external slowly time-varying wind (*with changing direction and speed*), a third experimental test is designed where the quadcopter is still forced to hover at a fixed point $\mathbf{p}_h$ in space, but the position of the mechanical fan with respect to $\mathbf{p}_h$ is forced to vary. Similar to the tests presented in Section VII-B, the following two experimental conditions are considered: before $t = t_c$, the fan is OFF; and (ii) from $t = t_c$ onward, the fan is ON. The experimental result, the time evolution of the position error, is shown in Fig. 9. In the presence of the time-varying wind, the RMSE and SD of $||\mathbf{z}_1||$, in steady state, are $0.044\,(\mathrm{m})$ and $0.014\,(\mathrm{m})$, respectively. Notice that these two error statistics are slightly larger than those obtained for constant disturbances (see Section VII-B, Fig. 7), but much smaller than what is observed without disturbance estimators (see Section VII-B, Fig. 8), attesting to the capability of slowly time-varying wind disturbance rejection.

D. Comparison With an Adaptive Strategy

To highlight the merit of the proposed estimation method (EM), we compare the performance of the proposed EM with that of the EM reported in [33]. Fig. 10 displays the time evolution of the mass estimation error $|\tilde{m}|$ for both methods under the presence of approximately constant external wind generated by a mechanical fan. The actual mass of the vehicle measured by an electronic scale is $200\,(\mathrm{g})$. The RMSE of $|\tilde{m}|$ obtained from the proposed EM $(2.2\,(\mathrm{g}))$ is much smaller than the value $(17.3\,(\mathrm{g}))$

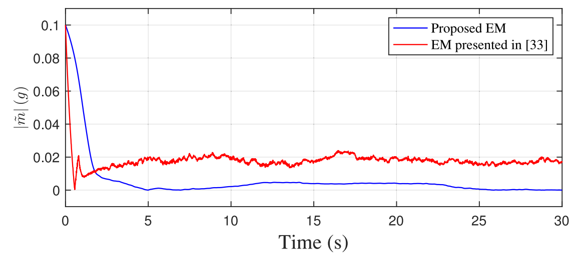

Fig. 10. Time evolution of the mass estimation error $|\tilde{m}|$ obtained using the proposed EM and the EM presented in [33] under the presence of constant external wind.

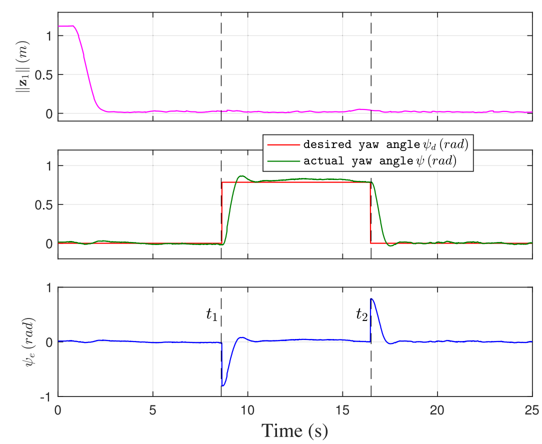

Fig. 11. Time evolution of the position error $\|\mathbf{z}_1\|$, desired yaw angle ψ_d $(0\,(\mathrm{rad})$ and $\pi/4\,(\mathrm{rad}))$, actual yaw angle ψ, and the yaw angle tracking error ψ_e.

obtained by using the EM proposed in [33], demonstrating the robustness and effectiveness of the proposed EM.

E. Yaw Angle Tracking Control Test

To validate the designed yaw control input ω_z, defined in (28), can enable the yaw to track the corresponding desired yaw angle, Fig. 11 is presented. During this test, the desired yaw angle is changed (at $t = t_1$ and $t = t_2$), while the vehicle is tracking the desired titled *eight-shape* path. From Fig. 11, it can be seen that the yaw tracking error ψ_e converges to a neighborhood of zero, whose RMSE in steady state is around $0.01\,(\mathrm{rad})$, justifying the efficacy of ω_z. The experimental test video, recorded from all tests, is presented in [41].

F. Comparison Analysis With the Existing Results

To further illustrate the effectiveness and performance of the proposed method, we compare our results with the experimental outcomes presented in [30], [32], and [42]. To be more specific, we make the following comparison analysis and conclusion.

1) Under the assumption that the quadrotor's mass and moment of inertia are known, a standard nonlinear path-following controller is developed in [30], where full nonlinear quadrotor models are used and global asymptotical stability (GAS) is achieved. Different from GAS, the proposed controller herein can only obtain GUUB, i.e.,

theoretically, we cannot drive the position error to zero. In this regard, GAS is better than GUUB. However, as we discussed in Section I, the presented control scheme in [30] needs one more backstepping iteration than the proposed method, which increases the order of the feedback system, resulting in a more complex controller with a relatively worse robustness. Notice that, in [30], the authors also provide experiments, showing that the maximum and RMSE of the position error are $8.0\,(\mathrm{cm})$ and $4.0\,(\mathrm{cm})$, respectively, which are larger than what we achieved, correspondingly, $5.0\,(\mathrm{cm})$ and $2.3\,(\mathrm{cm})$.

2) Adaptive methods, with less computations, are reported in [32] and [42]. Relying on a simplified quadrotor dynamic model (neglecting the cross coupling terms in the rotational movement), an adaptive tracking controller for a quadrotor is developed in [32], achieving asymptotic stability. Experiments presented therein display that the maximum position error in trajectory tracking is $10.0\,(\mathrm{cm})$, which is larger than $5.0\,(\mathrm{cm})$ obtained herein. A linear control technique is presented in [42], where the authors take into account the change of vehicle's mass and moment of inertia caused by the payload. However, the typical disadvantage of [42] is local stability, i.e., only when the system is close to the equilibrium, the system can provenly be stabilized.

To summarize, we conclude that a full nonlinear quadcopter model, which can better describe the system's characteristics (e.g., nonlinearity, strong coupling, and parameter uncertainties), is necessary for stabilizing a real quadcopter vehicle, which could lead to more complex computations in controller design. However, it helps to improve the tracking accuracy, robustness, and adaptivity. Additionally, it is possible to perform the resulting computations in real time even on a limited microcontroller due to the advances in computational capability.

VIII. Conclusion

Building on the backstepping technique, this article presented a control strategy to stabilize an underactuated quadcopter along a predefined path in the presence of uncertain vehicle mass, moment of inertia, and external disturbances. To attain robust stabilization, mass, external disturbances, and unknown constant terms associated to the moment of inertia estimators were designed, making it possible for a same quadcopter aircraft to adapt to different loads and maintain the performance, and with that ease the application of the proposed controller to a group of quadcopter aircrafts with different weights and sizes. Furthermore, the stability proof, simulation, and experimental results were presented, validating the performance of the proposed control strategy. In respect to the future work, it would include the following.

1) Designing time-varying disturbance observers for the vehicle to compensate for external time-varying disturbances.
2) Testing the developed controller in an outdoor environment where GPS and IMU are used to measure the states of the vehicle instead of the indoor motion capture system.

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